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Short Comunication

Application of Kruskal's Algorithm in Determining the Shortest Route Distance for Culinary Tourism in Mataram City

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Abstract. Mataram City as one of the tourist destination cities in West Nusa Tenggara has a variety of culinary attractions spread in various locations. The effectiveness of travel routes is an important factor in maximizing the culinary tourism experience, both in terms of travel time and cost efficiency. This research aims to apply Kruskal's Algorithm in determining the most effective culinary tourism travel route in Mataram City. Kruskal's algorithm, known as one of the methods in graph theory to find the minimum spanning tree (MST), is used to minimize the total travel distance between culinary location points. Location and distance data between culinary places were collected through field surveys and digital mapping. The results of the algorithm implementation show that the resulting route is able to connect all culinary points with a total travel distance that is more efficient than a random or conventional route of 15.09445 Km. Thus, the use of Kruskal Algorithm is proven effective as a solution in planning optimal culinary tourism routes in Mataram.

Keywords: Kruskal Algorithm, Culinary Tourism, Minimum Spanning Tree, Effective Route, Mataram City.

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1. Introduction

West Nusa Tenggara (NTB) Province, especially Lombok Island, is experiencing a significant acceleration of tourism development after the establishment of the Mandalika area as an international tourism Special Economic Zone (SEZ). One of the major momentum that drives a surge in tourist visits and global attention to NTB is the organization of MotoGP at Pertamina Mandalika International Circuit. The presence of this international event has a broad positive impact, including an increase in the number of tourist visits, infrastructure development, and the growth of the Mandalika tourism special economic zone (KEK) which is of international standard. Referring to the Mandalika 2022 MotoGP event, the event had an economic impact of IDR 3.57 trillion on the NTB economy and IDR 4.5 trillion on the national economy and absorbed a workforce of 4,600 people [1]. This prestigious international event not only increases the attractiveness of Lombok tourism but also opens up various opportunities for the development of supporting sectors, such as hospitality, transportation, creative economy, and culinary tourism. In line with the opinions of experts , Yakub states that emphasizes that tourism can increase foreign exchange earnings, create jobs, and stimulate the growth of the tourism industry so as to trigger economic growth [2]. Adyaharjanti, A., & Hartono stated that foreign tourist spending has a significant impact on the Indonesian economy. This shows that the tourism sector can be a catalyst for economic development [3].

Mataram City as the capital of NTB Province as well as the main entrance to Lombok has a vital role as a buffer city for Mandalika SEZ. Mataram's existence as an administrative, economic and social center makes it a strategic location in welcoming and serving the needs of tourists. One of Mataram's superior potentials that also becomes a magnet for tourists is the diversity of regional culinary tours. Local culinary or traditional food is a strong attraction and can improve the welfare of local residents. By enjoying local food, tourists have the opportunity to learn about the geography and culture of the local community and increase interest in visiting tourist destinations [4]. Specialties such as ayam taliwang, plecing kangkung, satay rembiga, and various other traditional snacks are a special attraction for tourists who want to experience the authentic culinary experience of Lombok [5]. However, the distribution of culinary tourism locations that are not centralized and scattered in various areas can be a challenge in itself, especially for tourists who want to enjoy culinary experiences efficiently in terms of time and distance. Without proper route planning, tourists risk spending a lot of time on the journey, which ultimately reduces the convenience and satisfaction of their visit. Therefore, there is a need for an approach that is capable of designing optimal and effective culinary travel routes.

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One solution that can be offered in answering these problems is through the application of algorithms in computer science and discrete mathematics, especially the Kruskal Algorithm. This algorithm is known as a method to find the Minimum Spanning Tree (MST) of a weighted graph, where all points (in this case culinary locations) can be connected with the minimum total weight (distance) without forming a cycle. According to some experts, MST is a fundamental and well-known optimization problem in graph theory [6]. MST is the determination of the edges connected between nodes in the network so that the minimum total side length is obtained [7], or a minimum cost tree is found, so as to decompose the problem logically and naturally smaller (Pop et al., 2018). Furthermore, according to Tamber et al (2020), MST is a tree formed from the edges given an undirected graph, with two properties, namely covering the graph of each vertex and providing the minimum value of the total of all edges can be as low as possible [8].

Kruskal algorithm is an algorithm in graph theory that finds a minimum spanning tree for a connected weighted graph [9]. Kruskal's algorithm works by selecting edges with the smallest weight that do not form a cycle, until all vertices are connected. The concept of tree graphs is very important in graph theory, especially in the context of graph theory applications. Trees play a crucial role in network analysis and design, often becoming the backbone of a connected network [10]. Since the definition of a tree is derived from graph theory, a tree can have only a vertex without an edge. Therefore, if $G = (V, E)$ is a tree, then V cannot be the empty set, but E can be empty. The application of Kruskal's algorithm is done by first sorting all the edges in the graph, then operating on them one by one until an n-1 spanning tree edge is reached (where n is the number of vertices in the graph [11]. More systematically, according to Seweken(2021), the Kruskal algorithm is an algorithm that can be used to find the minimum weight value of a spanning tree [12]. The following are the steps to find the MST using the Kruskal algorithm (1) Sort by the lowest weight value to the largest, (2) Select the edge with the smallest weight, (3) Select the edge with the smallest weight except the edge that has been selected, (4) Select the edge with the smallest weight but does not form a circuit with the vertex that has been marked, (5) Do the step repeatedly up to n-1 times. By applying Kruskal's Algorithm, it is possible to construct a culinary tour route that connects all locations in the network most efficiently.

Several studies that use the Kruskal algorithm in the network to find the shortest route that can be used as a reference in optimizing the minimum distance, namely research conducted by Erfariani on the Application of the Kruskal Algorithm in Determining the Distance of Effective Tourist Travel Routes in West Sumatra [13] then research conducted by Made Ayu Ulandari et al. on finding the shortest route to tourist attractions in Central Lombok [14], research on optimizing the shortest route of the internet network by Ar Ruhimat et al. [15], then about the shortest route of transportation routes by Dili et al. [16] then research on determining the shortest route of water distribution lines by Wulandari & Arifin [17]. From these studies, researchers were inspired to use the Kruskal algorithm in modeling the problem of optimizing culinary tourism travel routes in Mataram City. This is also a novelty in this research.

This research aims to apply the Kruskal Algorithm in determining the distance of effective culinary tourism travel routes in Mataram City. This research is also a real contribution in supporting the development of data and technology-based tourism infrastructure in the digital era. Furthermore, the results of this research are expected to provide route recommendations that can be utilized by related parties, both local governments, tourism businesses, and direct tourists, in order to improve the quality of tourism services and support the sustainable development of Mandalika SEZ as a national and international leading destination.

2. Method

This research is applied research and is a type of descriptive research [18]. Applied research is used to test the benefits of scientific theories and is focused on theoretical and practical knowledge in a particular field, not universal knowledge [19]. Applied research is directed at solving practical problems, in the context of determining policies [20]. The research was conducted in Mataram City, West Nusa Tenggara. The research object included 10 popular culinary tourism locations selected based on data from the NTB Tourism Office and reviews from online platforms such as Google Maps and TripAdvisor. These locations reflect Mataram's culinary diversity, such as Ayam Taliwang, Sate Rembiga, and Plecing Kangkung. This research restricts trips to be assumed to be made in one trip rather than returning to the hotel or place of lodging and then leaving again.

Data collection techniques are collected through: a) Field observations, namely using GPS to obtain geographical coordinates (latitude and longitude) of each culinary location. b) Distance measurement, namely using the Google Maps API to calculate the distance between locations in kilometers.

Data analysis was conducted through the following steps:

a) Graph Modeling: Each culinary location is represented as a node, and the distance between locations as an edge with weights in the form of distance traveled.

b) Application of Kruskal Algorithm:
   - Arrange all edges by weight from small to large.
   - Gradually add edges without forming a cycle until all vertices are connected.
   - The final result is a graph with minimum total weight (MST).

c) Graph Visualization: Using software to visualize the generated route.

The following is a flowchart of the steps to find the MST with the Kruskal algorithm:

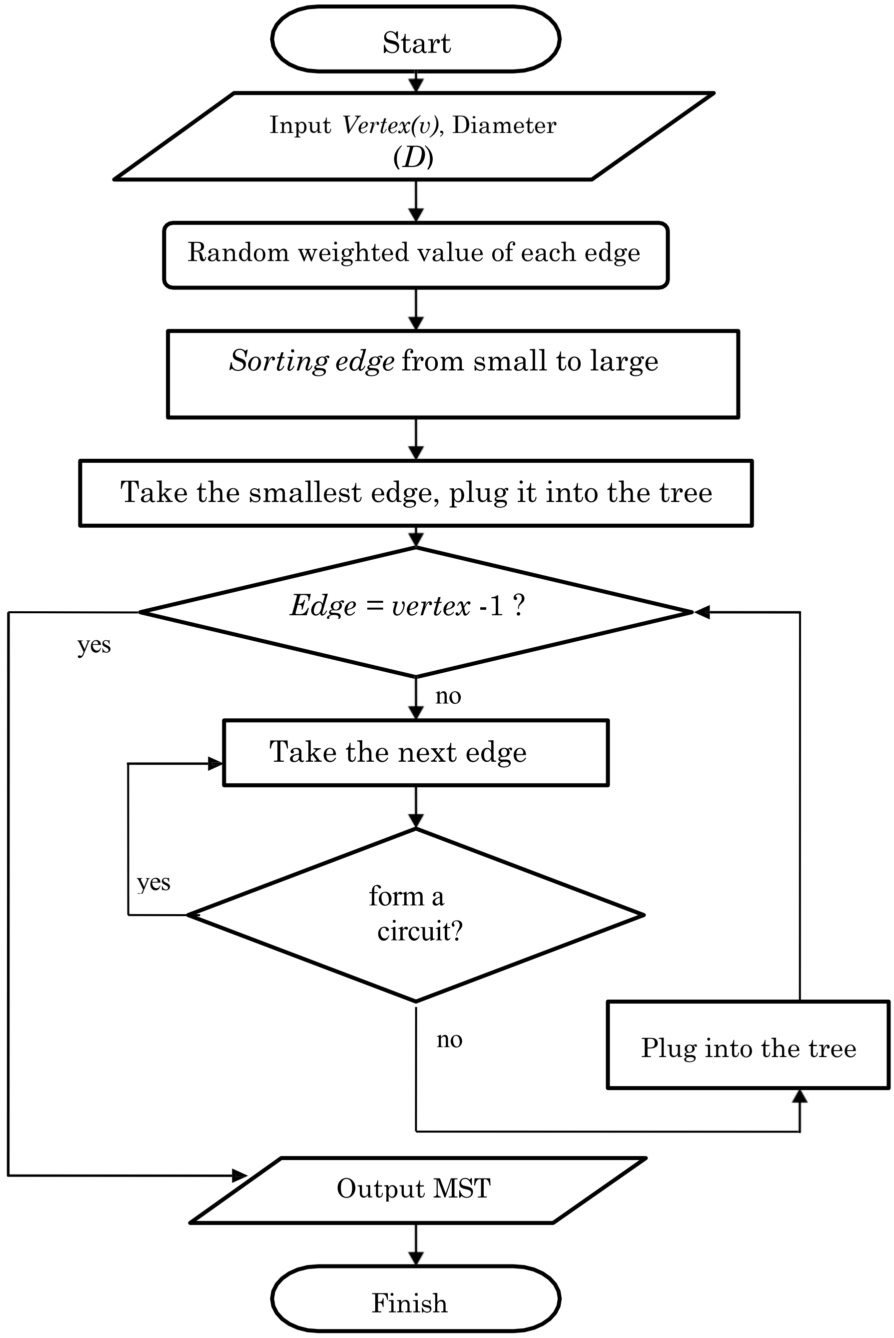

Figure 1. MST flowchart with Kruskal Algorithm

3. Result and Discussion

3.1 Graph Node Initiation

In this study, the first step we take is to initialize the vertices in the graph needed to distinguish between one vertex and another. The following is a list of names of culinary tourist attractions in Mataram City along with the initialization of the node names shown in Table 1 below:

Table 1. Name initiation Culinary tourist attractions

Place Name	Vertex (Vi)
Rumah Makan Khas Lombok	*A*
Roemah Langko	*B*
Taliwang Alam Nyaman	*C*
Sate Rembige Ibu Sinnaseh	*D*
Kuliner Bumigora Mataram	*E*
Lombok Kuliner	*F*
Taliwang Khas Pak Udin	*G*

In Table 1, the node mapping initiation is 7 node points. Determination of the node point is purely because the place is a favorite of tourists with a variety of mainstay menus served.

3.2 Location Map Modeling

Determination of the weight or distance between culinary tourist attractions in this study is the side of the graph that connects two vertices obtained from GPS distance on Maps. The following is a map of culinary tourist attractions in Mataram City using Google Maps as a reference in determining weights.

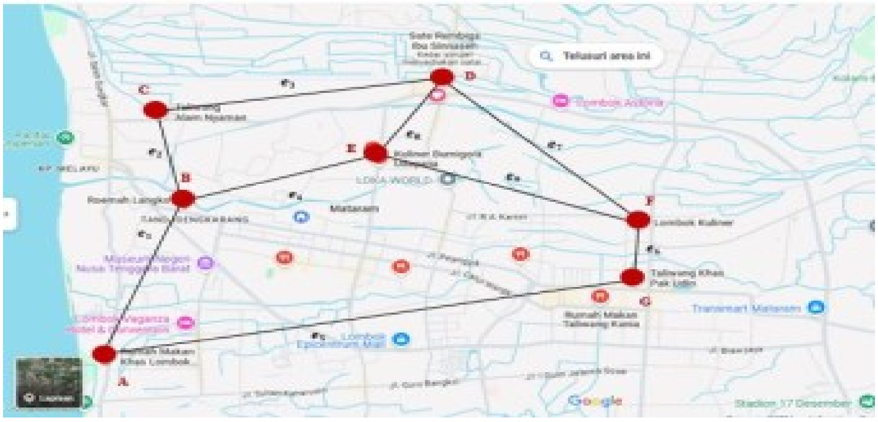

Figure 2. Plan of Culinary Tourist Attractions in Mataram City

In Figure 2, it can be seen that the plan has been given a name according to its node initiation. after that we can get the distance between places seen in Figure 3 below using Google Maps (in meters)

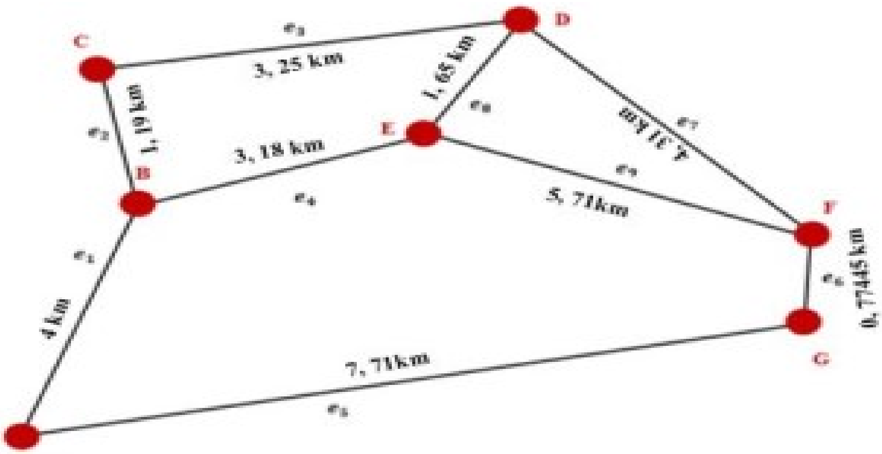

Figure 3. Weighted Graph (distance) based on Floor Plan

Based on Figure 3, 9 weighted edges are obtained in the following table 2:

Table 2. List of weights between vertices (edges)

Edges	*ei*	weighted/Distance (km)
A-B	e1	4
B-C	e2	1.19
C-D	e3	3.25
D-E	e8	1.65
B-E	e4	3.18
D-F	e7	4.31
E-F	e9	5.71
F-G	e6	0.77445
A-G	e5	7.71

In Table 2, it can be seen that the distance between the F-G vertices is the closest distance which is 0.77445 km while the farthest distance is the distance between the A-G vertices which is 7.71 km. After determining the amount of weight on the graph, the next step using Kruskal's algorithm is sorting in ascending order (from the smallest weight to the largest weight) so that the results in table 3 are obtained:

Table 3. Sorted list of weights from Smallest to Largest

Edges	ei	Weighted/Distance (km)
F-G	*e6*	0.77445
B-C	e2	1.19
D-E	e8	1.65
B-E	e4	3.18
C-D	e3	3.25
A-B	e1	4
D-F	e7	4.31
E-F	*e9*	5.71
A-G	*e5*	7.71

3.3 Application of Kruskal's algorithm

The application of Kruskal's algorithm begins by selecting the graph edge with the minimum distance weight, namely the F-G edge with a weight of 0.77445 and then we insert it into the following Figure 4. The application of Kruskal's algorithm in determining the shortest route distance begins by selecting the edge with the smallest weight, namely edge F–G with a distance of 0.77445 km, and inserting it into an initially empty graph. The process continues by selecting the next smallest edge, B–C, with a distance of 1.19 km, followed by edge D–E with a distance of 1.65 km. These edges are successively added to the spanning tree as long as they do not form any circuits. The fourth smallest edge, B–E, with a distance of 3.18 km, is then included, followed by edge C–D with a distance of 3.25 km. However, at this stage, the addition of edge C–D creates a cycle, which violates the fundamental rule of Kruskal's algorithm; hence, it is excluded from the tree. The process is then continued by selecting edge A–B, which has a distance of 4 km, and subsequently edge D–F with a distance of 4.31 km, both of which are added to the graph without forming a circuit. Edges E–F (5.71 km) and A–G (7.71 km) are evaluated but ultimately rejected because their inclusion would result in the formation of cycles. As Kruskal's algorithm is designed to avoid any circular connections within the graph, only acyclic connections are retained. Upon completion of the algorithm, the selected edges form a minimum spanning tree with a total distance of 15.09445 km, calculated by summing the distances of the accepted edges: 0.77445 + 1.19 + 1.65 + 3.18 + 4 + 4.31 km.

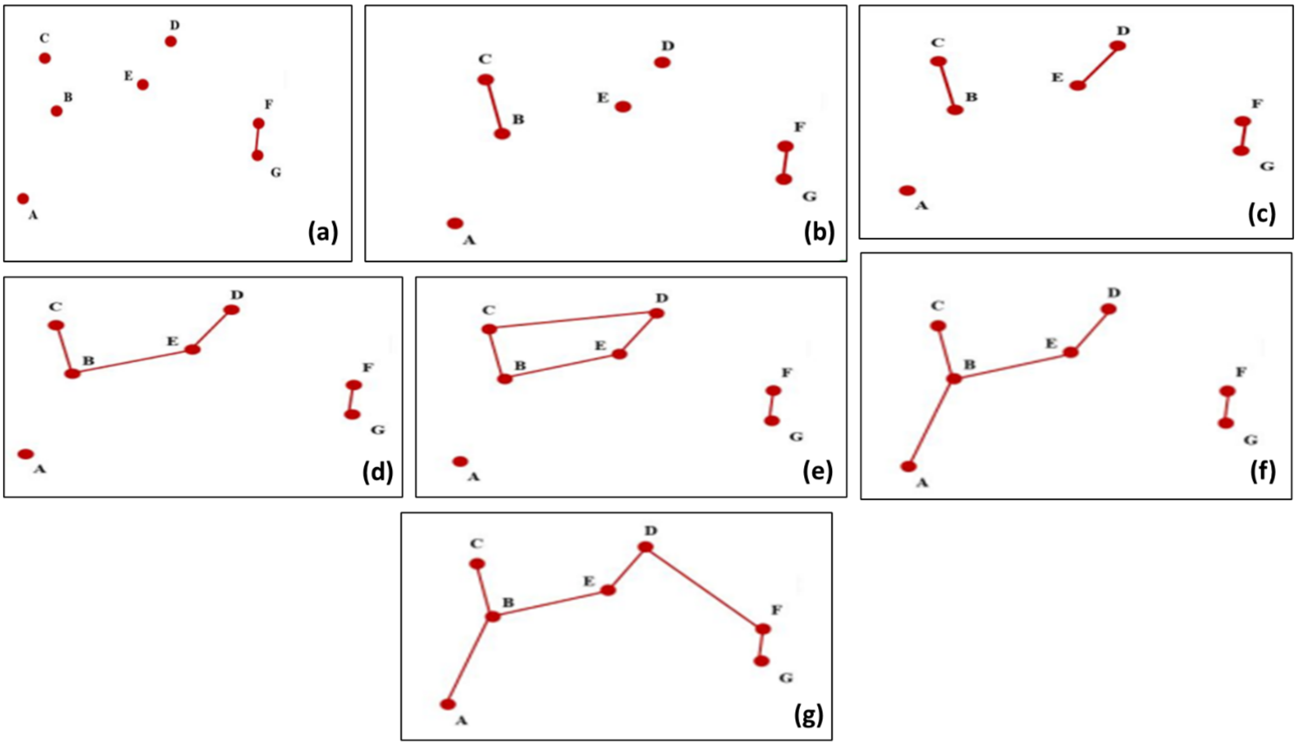

Figure 4. Stages of Kruskal's Algorithm in forming the minimum spanning tree for culinary tourism routes in Mataram City, starting from the lightest edge (a–g) and avoiding cycles to produce an optimal total distance.

4. Conclusion

Based on the research results, it is concluded that the Kruskal Algorithm can be used as a basis for determining the shortest route between culinary tourist attractions in Mataram City. Travel route with a distance of 15, 09445 Km. These results can be used as a reference for consideration of tourism actors in making culinary tourism travel routes in the city of Mataram. Researchers recommend further research using other methods or algorithms related to the minimum spanning tree.

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7. Author Declaration

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